Stable Sparse Pretraining

Deep Dive into Kimi K3 — Chapter 14

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Foundation Books Project

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Goal: type every route, owner, and endpoint; keep QB and MoonEP on separate clocks; return every partial to one history; keep values live through their last consumer; and recover only from a validated, coherent commit.

Roadmap

Two clocks, one route identity

One history per parameter

Liveness, checkpoints, and recovery

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Selection now, placement now, bias later



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- Fixture: semantic counts $(8, 4, 2, 2)$ stay fixed while rank loads go $(12, 4) \to (8, 8)$; outputs $(41, \ldots, 128)$ are identical under both placements.

One history per parameter

Repatriate, reduce, update once

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- Per-Head Muon normalizes each head block alone: perturbing head 1 leaves head 0 bit-for-bit unchanged; a stacked transform does not. [OBSERVED: book companion]

Liveness, checkpoints, and recovery

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Diagnose in dependency order. Next: Chapter 15, Long-Context Pretraining.